Course Project

Combined Heat and Power (CHP)

Course: REE-402 (Energy Efficiency)

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"Combining science, computation, and innovation for a sustainable future."

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Contents

List of Figures

List of Tables

Abstract

Combined Heat and Power (CHP) systems, also known as cogeneration, provide a highly efficient method for generating electricity and useful thermal energy from a single fuel source. Unlike conventional power plants that waste large quantities of heat, CHP systems recover and utilize this thermal energy for heating, industrial processes, or absorption cooling. This project presents a comprehensive engineering analysis of CHP technologies, operating principles, thermodynamic behavior, and performance metrics.

Mathematical models are provided to evaluate electrical efficiency, thermal efficiency, and overall system efficiency under varying operating conditions. A detailed CHP system flow diagram is included to illustrate the interaction between the prime mover, generator, heat recovery unit, and thermal load.

The report also explores the emerging concept of hybridizing CHP with solar flat-plate collectors and Phase Change Material (PCM) thermal storage. This configuration offers significant potential for reducing fossil fuel consumption and enhancing system flexibility. The findings highlight CHP as a reliable, cost-effective, and sustainable energy solution for modern distributed energy systems.

1 Introduction

Modern energy systems face increasing pressure to deliver reliable, affordable, and environmentally responsible power. Global electricity demand continues to rise due to rapid industrial growth, population expansion, and increased reliance on energy-intensive technologies. At the same time, concerns regarding greenhouse gas emissions, fuel price volatility, and energy security demand innovative solutions that maximize efficiency and reduce waste. Traditional centralized power plants typically convert only 30–40% of the chemical energy in fuel into electrical power. The remainder is lost as unused heat discharged into the atmosphere or cooling systems.

Combined Heat and Power (CHP) systems, also known as cogeneration, offer a highly effective alternative. CHP simultaneously generates electricity and useful thermal energy from a single fuel source, leading to total system efficiencies that can exceed 80%. The ability to recover and utilize waste heat significantly reduces primary fuel consumption and associated emissions, making CHP one of the most efficient and sustainable forms of distributed energy generation.

CHP systems are used in a wide range of applications, including industrial plants, district heating networks, hospitals, commercial buildings, and residential micro-CHP installations. Technological advancements in internal combustion engines, gas turbines, Stirling engines, and fuel cells have expanded the feasibility and flexibility of CHP across multiple sectors. Additionally, modern control systems, variable heat recovery units, and thermal storage technologies have improved operational performance, adaptability, and economic viability.

This project provides a comprehensive engineering analysis of CHP systems, focusing on their operating principles, thermodynamic performance, and practical implementation. The study discusses various prime movers, heat recovery configurations, efficiency metrics, and system design considerations. A detailed flow diagram is also provided to illustrate the energy pathways and major components of a CHP system.

In addition to conventional CHP concepts, this project explores a potential hybrid configuration combining CHP with solar flat-plate collectors and Phase Change Material (PCM) thermal storage. Such a system can reduce fossil fuel consumption, increase flexibility in meeting thermal loads, and support sustainability goals. This hybrid approach reflects the growing interest in integrating renewable energy with high-efficiency thermal systems.

The sections that follow present an organized and in-depth examination of CHP fundamentals, system components, thermodynamic analysis, performance evaluation, design considerations, case studies, and opportunities for future improvements.

2 Background

The global energy landscape is undergoing significant transformation due to increasing energy demand, sustainability goals, and pressure to reduce greenhouse gas emissions. Traditional power plants operate on the principle of converting fuel energy into electricity, but they typically achieve only 30–40% electrical efficiency. The remaining energy is lost as waste heat through exhaust gases, cooling towers, and other dissipation mechanisms. These losses contribute to higher operational costs, increased fuel consumption, and environmental impacts.

Combined Heat and Power (CHP), also known as cogeneration, addresses these inefficiencies by simultaneously generating electricity and useful thermal energy from a single fuel source. By capturing waste heat and redirecting it to supply heating, industrial processes, or hot water systems, CHP systems can reach total efficiencies of 70–90%. This makes CHP one of the most efficient forms of distributed energy generation currently available.

Historically, CHP technology dates back to early industrial facilities in the 19th century, where steam engines produced both mechanical power and process heat. Modern advancements in thermodynamics, materials, engine design, and automated control systems have expanded the applicability of CHP across many sectors. Today, CHP systems are widely used in universities, hospitals, district heating networks, chemical plants, food industries, commercial buildings, and residential micro-CHP installations.

Various prime movers can be used in CHP systems, including: - Internal combustion engines - Gas turbines - Microturbines - Steam turbines - Stirling engines - Fuel cells

The versatility of CHP allows it to operate on several fuel types such as natural gas, diesel, propane, landfill gas, and hydrogen-rich fuels. With the global shift toward cleaner energy, integrating CHP with renewable systems—such as solar thermal collectors or biomass gasifiers—offers additional environmental and economic advantages.

To illustrate the general structure and operation of a CHP system, Figure 1 shows a simplified flow diagram depicting the major components and energy pathways.

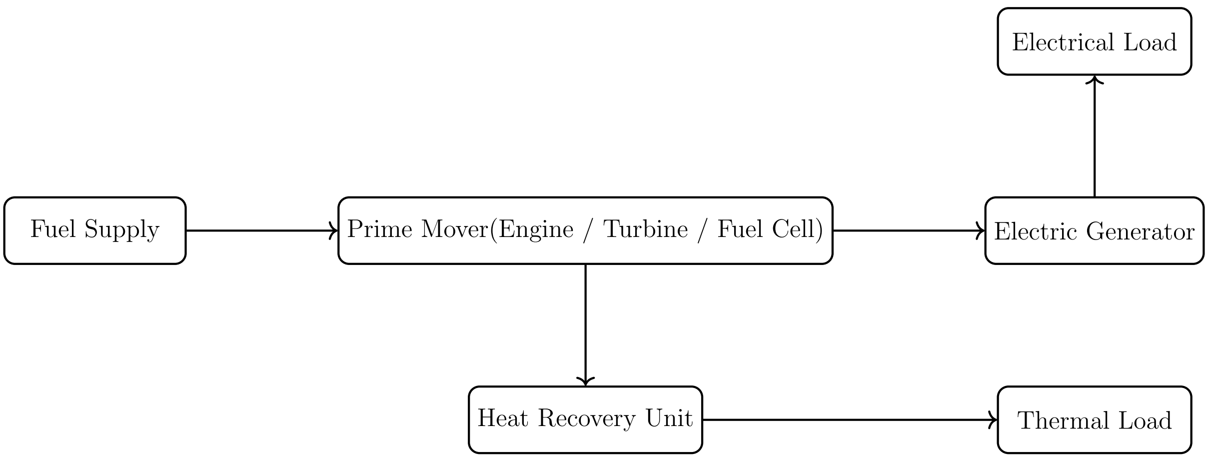

Figure 1: Flow diagram of a typical Combined Heat and Power (CHP) system.

3 CHP Principles and Thermodynamic Analysis

Combined Heat and Power (CHP) systems are based on the first and second laws of thermodynamics. The key idea is to convert the fuel chemical energy into useful work (electricity) and useful heat in a way that maximizes the total energy utilization and minimizes irreversibilities. In contrast to a conventional power plant, where almost all of the rejected heat is wasted to the environment, CHP systems reclaim a substantial portion of this heat to satisfy a local thermal demand.

3.1 Basic Energy Balance

For a steady-state CHP system, the overall energy balance can be written as

$$\dot{Q}_{\text{fuel}} = \dot{W}_{\text{el}} + \dot{Q}_{\text{useful}} + \dot{Q}_{\text{loss}}, \tag{1}$$

where

- $\dot{Q}_{\text{fuel}}$ is the rate of chemical energy input from the fuel,
- $\dot{W}_{\text{el}}$ is the net electrical power output,
- $\dot{Q}_{\text{useful}}$ is the rate of useful recovered heat delivered to the thermal load,
- $\dot{Q}_{\text{loss}}$ represents unavoidable losses to the environment.

In a conventional power-only plant, $\dot{Q}_{\text{useful}} \approx 0$, and almost all of the rejected heat appears in $\dot{Q}_{\text{loss}}$. In a CHP configuration, a large fraction of that heat is captured by the heat recovery unit and directed to a useful application.

3.2 Efficiency Definitions

To evaluate and compare CHP performance, several efficiency indicators are used.

3.2.1 Electrical efficiency.

$$\eta_{\text{el}} = \frac{\dot{W}_{\text{el}}}{\dot{Q}_{\text{fuel}}}. \tag{2}$$

This parameter measures how effectively the system converts fuel energy into electrical power. For engines and gas turbines in CHP mode, η_{el} typically ranges from 25% to 45%, depending on the technology and operating conditions.

3.2.2 Thermal efficiency.

$$\eta_{\text{th}} = \frac{\dot{Q}_{\text{useful}}}{\dot{Q}_{\text{fuel}}}. \tag{3}$$

This describes the fraction of fuel energy recovered as useful heat for heating, hot water, or process loads.

3.2.3 Total (overall) CHP efficiency.

$$\eta_{\mathrm{CHP}} = \frac{\dot{W}_{\mathrm{el}} + \dot{Q}_{\mathrm{useful}}}{\dot{Q}_{\mathrm{fuel}}} = \eta_{\mathrm{el}} + \eta_{\mathrm{th}}. \tag{4}$$

For a well-designed CHP system, η_{CHP} can reach 70–90%, significantly higher than the 30–40% typical of power-only plants.

3.2.4 Power-to-heat ratio. Another important indicator is the electrical-to-thermal output ratio:

$$\Phi_{P/H} = \frac{\dot{W}_{\mathrm{el}}}{\dot{Q}_{\mathrm{useful}}}. \tag{5}$$

This ratio helps match the CHP system to the local demand profile. Prime movers with a high $\Phi_{P/H}$ are better suited for applications where electricity demand dominates, while low $\Phi_{P/H}$ units are preferable for heat-dominated sites such as district heating networks.

3.2.5 Primary energy saving (PES). To quantify the benefit of CHP compared with separate generation of electricity and heat, the primary energy saving index is defined as

$$\mathrm{PES} = 1 - \frac{\dot{Q}_{\mathrm{fuel,CHP}}}{\dfrac{\dot{W}_{\mathrm{el}}}{\eta_{\mathrm{el,ref}}} + \dfrac{\dot{Q}_{\mathrm{useful}}}{\eta_{\mathrm{boiler,ref}}}}, \tag{6}$$

where $\eta_{\mathrm{el,ref}}$ is the efficiency of a reference power plant and $\eta_{\mathrm{boiler,ref}}$ is the efficiency of a reference boiler. Positive PES values indicate that the CHP system consumes less primary energy than the separate generation alternative.

3.3 Illustrative Numerical Example

Consider a gas-engine-based CHP unit that consumes a fuel input of $\dot{Q}_{\mathrm{fuel}} = 1000$ kW. The system delivers $\dot{W}_{\mathrm{el}} = 350$ kW of electricity and $\dot{Q}_{\mathrm{useful}} = 450$ kW of useful thermal energy.

Using Equations (2)–(4):

$$\eta_{\mathrm{el}} = \frac{350}{1000} = 0.35\ (35\%), \tag{7}$$

$$\eta_{\mathrm{th}} = \frac{450}{1000} = 0.45\ (45\%), \tag{8}$$

$$\eta_{\mathrm{CHP}} = 0.35 + 0.45 = 0.80\ (80\%). \tag{9}$$

If the same electricity were produced in a conventional power plant with $\eta_{\mathrm{el,ref}} = 0.38$, and the heat supplied by a boiler with $\eta_{\mathrm{boiler,ref}} = 0.90$, the separate generation would require

$$\dot{Q}_{\mathrm{fuel,sep}} = \frac{\dot{W}_{\mathrm{el}}}{\eta_{\mathrm{el,ref}}} + \frac{\dot{Q}_{\mathrm{useful}}}{\eta_{\mathrm{boiler,ref}}} \tag{10}$$

$$= \frac{350}{0.38} + \frac{450}{0.90} \approx 921 + 500 = 1421 \text{ kW}. \tag{11}$$

Therefore, the primary energy saving of the CHP system is

$$\mathrm{PES} = 1 - \frac{1000}{1421} \approx 0.296 \ (29.6\%), \tag{12}$$

which shows that the CHP configuration reduces primary energy consumption by almost 30% compared to separate generation.

3.4 Operating Modes

CHP systems can be operated under different control strategies, mainly:

- **Heat-led operation:** the CHP follows the thermal demand. Electricity is a by-product, and any deficit or surplus is covered by the grid.
- **Power-led operation:** the CHP follows the electrical demand. Excess or deficit heat is handled by auxiliary boilers or thermal storage.
- **Balanced operation with thermal storage:** the CHP is operated near its optimal efficiency point while a storage tank absorbs mismatches between generation and demand.

These operating modes are important when integrating CHP with other technologies, such as solar collectors and PCM-based thermal storage, which will be discussed in later sections.

4 CHP Technologies and Configurations

A variety of prime mover technologies can be used in CHP systems. The selection of a suitable technology depends on the electrical and thermal demand profile, fuel type, required reliability, environmental regulations, and economic constraints. This section provides an overview of the most common CHP technologies and highlights their typical operating characteristics.

4.1 Internal Combustion Engines

Reciprocating internal combustion engines are widely used in small and medium-size CHP applications, typically ranging from a few kilowatts up to several megawatts. Spark-ignition gas engines operate mainly on natural gas, biogas, or landfill gas, while compression-ignition engines are commonly fueled by diesel or heavy fuel oils.

The main advantages of engine-based CHP systems include:

- High electrical efficiency in the range of 30–45%.
- Fast start-up and good load-following capability.
- Mature technology with well-established maintenance practices.
- Suitability for modular multi-unit installations.

Waste heat is recovered from the engine exhaust gases and from the jacket-water cooling system. The combined recovery can provide thermal energy at temperatures between 80 °C and 450 °C, suitable for space heating, domestic hot water, and some low- to medium-temperature industrial processes.

4.2 Gas Turbines

Gas turbines are typically used for larger CHP capacities, from around 1 MW up to more than 100 MW. They operate on the Brayton cycle, where compressed air is mixed with fuel, combusted at nearly constant pressure, and expanded through a turbine that drives both the compressor and the electric generator.

Gas turbines exhibit the following characteristics:

- Electrical efficiencies typically between 25–38% for simple-cycle units.
- High-temperature exhaust gases (400–550 °C) well suited for heat recovery.
- Lower specific emissions of NO_x and particulates compared with many reciprocating engines.
- Best efficiency at high and relatively constant loads.

The exhaust gases are directed to a heat recovery steam generator (HRSG) or to a direct heat exchanger. In some configurations, a secondary steam turbine is added, forming a combined-cycle CHP system with very high overall efficiency.

4.3 Microturbines

Microturbines are small gas turbines with power ratings typically in the range of 30–300 kW. They are compact, have few moving parts, and can operate on a variety of gaseous fuels.

Key features of microturbine CHP systems include:

- Electrical efficiency around 25–32%.
- High exhaust gas temperature (250–300 °C) that can be recovered for low- to medium-temperature heating.
- Low vibration and noise levels.
- Reduced maintenance requirements due to air bearings and simple construction.

Microturbines are particularly suitable for commercial buildings, small industrial facilities, and district energy systems where a relatively continuous thermal demand exists.

4.4 Steam Turbines

Steam turbine-based CHP plants are commonly found in industrial sites and district heating networks. In such systems, a boiler or HRSG produces high-pressure steam, which is expanded through a turbine to generate electricity. Partially expanded steam is then extracted at an intermediate pressure and used to satisfy the process or heating loads.

Steam turbines are characterized by:

- High reliability and long service life.
- Capability to utilize a wide range of fuels, including coal, biomass, waste-derived fuels, and process by-products.
- Relatively low electrical efficiency for small units, with better performance at larger scale.

Back-pressure steam turbines are commonly used when there is a large, continuous process steam demand. Extraction-condensing turbines can flexibly adjust the split between electricity and heat production.

4.5 Stirling Engines

Stirling engines are external combustion engines that can operate on many fuel types or even concentrated solar energy. Although still less common in commercial CHP installations, they offer several attractive features:

- Very low noise and vibration levels.
- Potentially high efficiency at small scales.
- Fuel flexibility, including biomass and solar thermal input.

Micro-CHP Stirling engines for residential applications are typically in the range of 0.5–10 kW of electrical output, with a significantly higher thermal output used for space and water heating.

4.6 Fuel Cell CHP Systems

Fuel cells convert the chemical energy of hydrogen-rich fuels directly into electricity and heat through electrochemical reactions. Several fuel cell types are suitable for CHP, such as phosphoric acid fuel cells (PAFC), molten carbonate fuel cells (MCFC), and solid oxide fuel cells (SOFC).

Advantages of fuel-cell-based CHP include:

- High electrical efficiency, often in the range of 40–55%.
- Very low pollutant emissions and quiet operation.
- Modular design, suitable for distributed generation and microgrids.

The waste heat from fuel cells is typically available at temperatures between 60 °C and 250 °C, depending on the technology. This heat can be used for space heating, hot water, or absorption cooling.

4.7 Comparison of Technologies

Table 1 qualitatively compares the main CHP technologies in terms of size range, electrical efficiency, and typical applications.

Table 1: Qualitative comparison of common CHP prime movers.

Technology	Power Range	El. Efficiency	Typical Applications
Reciprocating engine	10 kW–10 MW	30–45%	Buildings, industry, microgrids
Gas turbine	1 MW–100 MW	25–38%	Industry, district heating
Microturbine	30–300 kW	25–32%	Commercial buildings, small industry
Steam turbine	>1 MW	15–35%	Large industry, district heating
Stirling engine	0.5–10 kW	15–30%	Residential micro-CHP
Fuel cell	5 kW–5 MW	40–55%	High-efficiency distributed generation

In practice, the optimal choice of technology depends on matching the CHP system to the local demand profile, fuel availability, environmental regulations, and investment constraints.

5 Integration of CHP with Solar Flat-Plate Collectors and PCM Storage (Future Work)

While conventional CHP systems already offer significant efficiency and environmental benefits, further improvements are possible through hybridization with renewable energy and advanced thermal storage. This section outlines a conceptual design in which a CHP unit is combined with a solar flat-plate collector field and a Phase Change Material (PCM) storage tank. The goal is to reduce fossil fuel consumption, increase flexibility in meeting thermal loads, and enhance overall system resilience.

5.1 Conceptual System Description

In the proposed hybrid configuration, the CHP unit remains the primary source of electricity and high-grade heat. A solar thermal subsystem based on flat-plate collectors is added to supply additional low- to medium-temperature heat. Excess solar energy is stored in a PCM tank, which can later discharge heat when solar input is low or when the CHP unit is operating at part load.

At a conceptual level, the main components of the integrated system are:

- A prime mover (e.g., gas engine or microturbine) and generator.
- A heat recovery unit connected to the CHP exhaust and cooling circuits.
- A hot-water distribution loop supplying the building or industrial thermal load.
- A solar flat-plate collector array connected to the same thermal loop.
- A PCM thermal storage tank integrated in parallel or series with the hot-water tank.
- Control valves and pumps to manage charging and discharging of the storage.

In future work, a detailed flow diagram for this hybrid system can be developed in TikZ, showing the interactions between CHP, solar collectors, and PCM storage.

5.2 Solar Flat-Plate Collector Subsystem

Flat-plate collectors are widely used for solar water heating and low-temperature thermal applications. They consist of an absorber plate, transparent cover, insulation, and a network of tubes or channels through which a working fluid circulates.

For integration with CHP, the solar collectors are typically connected to the return line of the heating circuit. The collector outlet temperature is chosen so that it can either:

- Directly contribute to the building heating load, or
- Charge the PCM tank when the thermal demand is partially satisfied by the CHP unit.

The useful heat gain from the collector field can be expressed as

$$\dot{Q}_{\mathrm{sol}} = \dot{m}_{\mathrm{sol}} c_p (T_{\mathrm{out}} - T_{\mathrm{in}}), \tag{13}$$

where $\dot{m}_{\mathrm{sol}}$ is the mass flow rate through the collector loop, c_p is the specific heat capacity of the fluid, and T_{in} and T_{out} are the inlet and outlet temperatures, respectively.

5.3 PCM Thermal Storage Tank

Phase Change Materials provide high energy storage density by exploiting the latent heat associated with melting and solidification. For building heating applications, suitable PCM materials often have phase-change temperatures in the range of 35–70 °C.

The PCM tank can be integrated in different ways:

- **Series configuration:** the water passes through the PCM tank before reaching the load, allowing the tank to discharge and smooth temperature fluctuations.
- **Parallel configuration:** the PCM tank is charged during periods of surplus thermal energy and discharged when additional heat is required.

The amount of thermal energy stored in the PCM can be approximated by

$$Q_{\mathrm{PCM}} \approx m_{\mathrm{PCM}} \left[c_{p,s} (T_{\mathrm{m}} - T_{\mathrm{min}}) + h_f + c_{p,l} (T_{\mathrm{max}} - T_{\mathrm{m}}) \right], \tag{14}$$

where m_{PCM} is the PCM mass, $c_{p,s}$ and $c_{p,l}$ are the specific heats in solid and liquid phases, h_f is the latent heat of fusion, T_{m} is the melting temperature, and T_{min}, T_{max} define the useful operating range.

5.4 Operating Strategies for the Hybrid System

Several operating strategies can be proposed for the integrated CHP–solar–PCM system:

- **Solar-priority mode:** when solar radiation is available, the flat-plate collectors and PCM storage are used to satisfy as much of the thermal load as possible. The CHP unit is operated at reduced output or shut down if electrical demand is low.
- **CHP-priority mode:** during periods of high electrical demand or low solar input, the CHP system operates near its optimal efficiency point. Any surplus heat is used to charge the PCM tank.
- **Balanced mode with storage:** control algorithms coordinate CHP and solar operation such that the PCM tank absorbs short-term mismatches between thermal generation and demand.

These strategies can be evaluated using dynamic simulation tools to determine the impact on fuel consumption, renewable energy fraction, storage utilization, and operating costs.

5.5 Expected Benefits and Challenges

The main expected benefits of integrating CHP with solar collectors and PCM storage include:

- Reduced annual fuel consumption and operating cost.
- Increased share of renewable energy in the overall heat supply.
- Improved flexibility and load-following capability due to thermal storage.
- Potential reduction of start–stop cycling of the CHP unit.

However, several challenges must also be considered:

- Additional capital cost for solar collectors, PCM materials, and storage tanks.
- Integration complexity and the need for advanced control strategies.
- PCM performance degradation over many melting/solidification cycles.
- Space constraints for installing both collectors and storage tanks.

5.6 Proposed Future Work

For this course project, a detailed future work plan could include:

- Developing a full schematic flow diagram of the integrated CHP–solar–PCM system.
- Selecting a specific PCM material and designing the storage tank size based on typical load profiles.
- Building a simplified mathematical model to compare fuel consumption and primary energy savings between:
    1. CHP-only operation,
    2. CHP + solar collectors, and
    3. CHP + solar collectors + PCM storage.
- Performing a parametric study on key variables such as solar collector area, PCM mass, and CHP operating strategy.

These steps would extend the current project into a comprehensive analysis of hybrid CHP systems and provide a strong foundation for a future senior design or research thesis.

6 Future Work

The analysis presented in this report shows that Combined Heat and Power (CHP) systems can significantly improve overall energy efficiency by simultaneously generating electricity and useful heat. However, there remains considerable potential to further enhance performance and sustainability by integrating CHP with renewable solar thermal technologies and advanced thermal energy storage (TES) based on phase change materials (PCMs). This section outlines several directions for future work, with particular focus on coupling a solar flat-plate collector field and a PCM storage tank—similar to the configuration used in the author's senior design project—with the CHP concept studied in this course project.

6.1 Solar-Assisted CHP Using Flat-Plate Collectors and PCM Storage

A promising extension of the present work is the development of a solar-assisted CHP system in which a flat-plate solar collector array preheats the working fluid before it enters the CHP heat recovery unit or directly charges a PCM-based hot water storage tank. During periods of high solar irradiance, the collectors would supply part of the thermal demand for space heating and domestic hot water, thereby reducing the heat that must be provided by the CHP unit. This would allow the prime mover to operate closer to its optimum electrical efficiency point while still satisfying the building thermal load.

The PCM storage tank from the senior design project can be used as a buffer between the variable solar resource and the relatively smoother thermal load profile. When solar availability is high and thermal demand is low, surplus solar energy can be stored in the PCM tank in the form of latent heat at nearly constant temperature. During cloudy periods, at night, or when the electrical demand is high but the CHP exhaust heat alone is insufficient, the stored thermal energy can be discharged to support the heating system. This hybrid configuration would enable higher renewable contribution, improved load matching, and reduced fuel consumption for the CHP unit.

Future work should begin with a conceptual flow diagram of the integrated system, showing the flat-plate collectors, solar circulation loop, PCM tank, CHP heat recovery unit, auxiliary boiler (if any), and the heating distribution network. Particular attention should be given to the hydraulic arrangement—series, parallel, or mixed configurations between the solar loop, CHP heat recovery, and storage tank—and to the placement of three-way valves and pumps that enable different operating modes such as 'solar-only," 'CHP-only," and "solar + CHP + storage".

6.2 Electric PCM Storage Heater Concept

In addition to integrating PCM storage with CHP and solar thermal systems, a related concept for future work is the use of a PCM storage tank with an external electric heating coil. In this configuration, the tank is charged using electricity (for example, from the grid, photovoltaic panels, or a CHP unit), and the stored thermal energy is later used to supply space heating and domestic hot water in residential or commercial buildings. Compared with a conventional electric heater, this approach can shift part of the electrical demand to off-peak periods and reduce peak loads while providing more stable outlet temperatures.

Figure 6.2 illustrates the main components and energy flows of the proposed electric PCM storage heater.

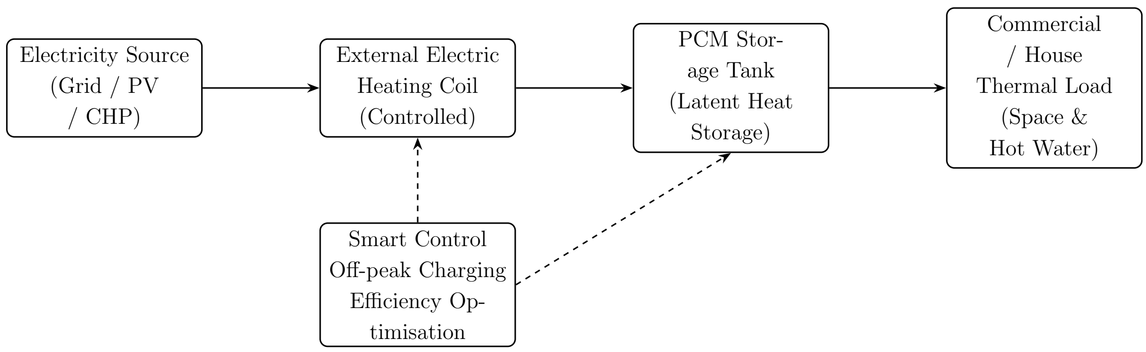

Figure 2: Conceptual flow diagram of the proposed electric heater with PCM storage tank and external coil for residential and commercial applications.

6.3 Thermal Design and Sizing of the Solar–PCM Subsystem

Another important task for future work is the detailed thermal design of the solar–PCM subsystem. The following design variables should be investigated:

- Solar collector aperture area and tilt angle, optimized for the local climate and the target load (space heating, domestic hot water, or both).
- PCM melting/solidification temperature range, chosen to match the typical supply and return temperatures of the heating system and the exhaust-gas temperature from the CHP heat recovery unit.
- Mass of PCM and storage tank volume required to cover a selected fraction of the daily or weekly thermal demand.
- Heat transfer enhancement techniques inside the PCM tank (fins, encapsulation geometry, internal heat exchangers) to improve charge–discharge rates.

Parametric simulations can be carried out to explore the relationship between these design parameters and key performance indicators such as solar fraction, CHP fuel savings, peak-load reduction, and number of charge–discharge cycles per season. The PCM tank design should build on established models for latent TES, including both sensible and latent heat contributions and the effect of partial melting/solidification.

6.4 Control Strategies for Hybrid CHP–Solar–PCM Operation

The effectiveness of the hybrid system will strongly depend on the control strategy. The PCM thesis used in this work shows that simple time-based control of PCM charging and discharging can lead to sub-optimal utilization, and that more advanced strategies based on temperature measurements, flow control, and state-of-charge (SoC) estimation can significantly improve performance. Inspired by those findings, future work on the CHP–solar–PCM system should consider:

- Temperature-based logic for when to charge the PCM tank from solar collectors, from CHP exhaust heat, or from both.
- Prioritization rules that decide whether heat from the CHP unit should first serve the load directly, charge the PCM tank, or be bypassed when the tank is full and demand is low.
- Variable flow-rate control for the solar loop and PCM charging circuit to optimize heat transfer while minimizing pump energy consumption.
- Model-based SoC estimation for the PCM tank, using inlet/outlet temperatures and flow measurements to determine available storage capacity and to prevent overcharging or incomplete solidification.

Advanced control approaches, such as rule-based supervisory control or model predictive control, could be investigated to coordinate the operation of the CHP, solar collectors, and PCM tank. The objective functions may include minimization of fuel consumption, operating cost, or CO_2 emissions, while satisfying comfort constraints and equipment limitations.

6.5 Dynamic Modelling and Simulation of the Integrated System

To properly evaluate the dynamic interaction between the CHP unit, solar collectors, PCM storage, and building loads, it is recommended to develop a detailed transient simulation model of the combined system. Following the methodology used in the PCM storage thesis, the model could be implemented in a multi-domain environment such as Modelica/Dymola, TRNSYS, or MATLAB/Simulink, with component models for the prime mover, heat recovery unit, flat-plate collectors, pumps, valves, and the PCM tank.

The PCM tank model should capture phase change behaviour, heat transfer coefficients, and stratification effects. The solar collector model should include transient response to irradiance and ambient temperature. By simulating the system over representative days and seasons, it will be possible to quantify:

- Seasonal energy balances for electricity, useful heat, and stored/recovered thermal energy.
- The impact of different control strategies on CHP operating hours, start–stop cycles, and part-load efficiency.
- Peak-load shaving and reduction in auxiliary boiler or electric heater usage.
- Sensitivity to climate conditions, load profiles, and component sizing.

Such a model will also serve as a platform for testing alternative configurations (for example, using the PCM tank for cooling, or coupling it with an absorption chiller) without the cost and risk of physical experiments.

6.6 Experimental Validation and Performance Monitoring

Where possible, future work should include experimental validation of the proposed hybrid system. This could be done at laboratory scale or through a pilot installation that combines a small CHP unit with a flat-plate solar collector array and a PCM storage tank similar to the senior design prototype. Key measurements would include temperatures, flow rates, electric and thermal outputs, and solar irradiance.

Long-term monitoring would make it possible to evaluate system reliability, degradation of PCM properties over repeated cycles, and real-world control performance. Measured data could then be used to refine the simulation models, validate SoC algorithms, and verify predicted fuel savings and emission reductions.

6.7 Economic and Environmental Assessment

Finally, to support real-world adoption, future work should also address the techno-economic and environmental assessment of the hybrid CHP–solar–PCM system. This includes:

- Life-cycle cost analysis comparing the integrated system with conventional CHP-only and boiler-based systems.
- Estimation of payback period, net present value, and levelized cost of energy under different fuel prices and incentive schemes.
- Evaluation of CO_2 emission reductions and primary energy savings for various renewable penetration levels and operating strategies.

Combining rigorous technical modelling with economic and environmental assessment will provide a comprehensive picture of the potential benefits of integrating flat-plate solar collectors and PCM thermal storage with CHP systems. This future work can form the basis for a full research project or a subsequent senior or master thesis, and it directly extends both the current CHP course project and the previous flat-plate–PCM design experience of the author.

7 Case Study: Example CHP System

To further illustrate the engineering concepts discussed in this report, this section presents a simplified case study of a building-scale CHP installation. The objective is not to design a fully optimized plant, but rather to apply the performance indicators introduced earlier and to highlight the potential benefits of CHP compared with conventional separate generation of heat and electricity.

7.1 Description of the Reference Building

Consider a medium-size educational building, such as a university department block, located in a region with a relatively cool climate. The building is assumed to have a peak electrical demand of 300 kW and a peak thermal demand of 500 kW for space heating and domestic hot water during the winter season. The annual electricity consumption is estimated at 1.2 GWh, and the annual heat demand at 1.8 GWh.

In a conventional configuration, electricity is purchased from the grid and all heat is supplied by a natural-gas-fired boiler with an efficiency of about 90%. The electricity mix of the grid has an average primary energy efficiency of 38% and a certain CO_2 emission factor, which will be used for comparison.

7.2 Proposed CHP Configuration

For the CHP case, a gas-engine-based cogeneration unit with a nominal electrical capacity of 200 kW and a corresponding thermal output of 250 kW is installed. The unit is sized to cover the base thermal load of the building, while peak heat demand is still met by the existing boiler. The key performance data of the CHP unit are assumed as follows:

- Fuel input at full load: 600 kW (LHV basis).
- Electrical efficiency: $\eta_{\mathrm{el}} = 0.33$.
- Thermal efficiency: $\eta_{\mathrm{th}} = 0.42$.
- Overall efficiency: $\eta_{\mathrm{CHP}} = 0.75$.

The CHP unit is operated in heat-led mode. During the heating season it runs for approximately 4 000 hours per year at an average load of 80%, supplying a large fraction of the building's base electrical and thermal demands.

7.3 Annual Energy Balances

At 80% load the average fuel input to the CHP unit is

$$\dot{Q}_{\text{fuel,avg}} = 0.8 \times 600 \text{ kW} = 480 \text{ kW}. \tag{15}$$

The corresponding electrical and thermal outputs are

$$\dot{W}_{\text{el,avg}} = \eta_{\text{el}} \dot{Q}_{\text{fuel,avg}} = 0.33 \times 480 \approx 158 \text{ kW}, \tag{16}$$
$$\dot{Q}_{\text{th,avg}} = \eta_{\text{th}} \dot{Q}_{\text{fuel,avg}} = 0.42 \times 480 \approx 202 \text{ kW}. \tag{17}$$

Over 4 000 hours of operation per year, the annual energy outputs are therefore

$$E_{\text{el,CHP}} \approx 158 \times 4000 = 632 \text{ MWh}, \tag{18}$$
$$E_{\text{th,CHP}} \approx 202 \times 4000 = 808 \text{ MWh}. \tag{19}$$

The building still needs additional electricity and heat from the grid and boiler, respectively. The remaining annual demands are

$$E_{\text{el,grid}} = 1\,200 - 632 = 568 \text{ MWh}, \tag{20}$$
$$E_{\text{th,boiler}} = 1\,800 - 808 = 992 \text{ MWh}. \tag{21}$$

Assuming a boiler efficiency of 90%, the boiler fuel consumption is

$$E_{\text{fuel,boiler}} = \frac{E_{\text{th,boiler}}}{0.90} \approx \frac{992}{0.90} \approx 1\,102 \text{ MWh}. \tag{22}$$

The CHP fuel consumption is

$$E_{\text{fuel,CHP}} = \dot{Q}_{\text{fuel,avg}} \times 4000 = 480 \times 4000 = 1\,920 \text{ MWh}. \tag{23}$$

Thus, the total primary energy use in the CHP case is

$$E_{\text{fuel,total,CHP}} = 1\,920 + 1\,102 = 3\,022 \text{ MWh}. \tag{24}$$

7.4 Comparison with Separate Generation

For separate generation, the electricity is supplied entirely by the grid and the thermal load by the boiler. The corresponding fuel use is

$$E_{\text{fuel,el,sep}} = \frac{E_{\text{el,total}}}{\eta_{\text{el,ref}}} = \frac{1\,200}{0.38} \approx 3\,158 \text{ MWh,} \tag{25}$$

$$E_{\text{fuel,th,sep}} = \frac{E_{\text{th,total}}}{\eta_{\text{boiler,ref}}} = \frac{1\,800}{0.90} \approx 2\,000 \text{ MWh.} \tag{26}$$

So the total primary energy consumption without CHP is

$$E_{\text{fuel,total,sep}} = 3\,158 + 2\,000 = 5\,158 \text{ MWh.} \tag{27}$$

The primary energy saving due to CHP can be estimated as

$$\text{PES} = 1 - \frac{E_{\text{fuel,total,CHP}}}{E_{\text{fuel,total,sep}}} = 1 - \frac{3\,022}{5\,158} \approx 0.414 \text{ (41.4\%).} \tag{28}$$

This simple case study indicates that, for the assumed operating conditions, the CHP system could reduce primary fuel consumption by more than 40% compared with the reference system using separate generation of heat and electricity.

7.5 Qualitative Environmental and Economic Benefits

If the fuel used for both the CHP unit and the boiler is natural gas, the reduction in primary energy use will translate directly into proportional reductions in CO_2 emissions. In addition, the CHP unit displaces electricity from the grid, which may have a higher emission factor if it includes coal- or oil-fired generation. The combined effect is an even larger emission reduction at the system level.

From an economic perspective, the benefits arise from lower fuel consumption and reduced purchases of grid electricity, balanced against the capital cost and maintenance cost of the CHP installation. A detailed financial analysis would require local fuel and electricity prices, investment cost data, and assumptions about discount rate and project lifetime. Nevertheless, the case study illustrates that, under suitable conditions, CHP can provide both environmental and economic advantages.

8 Discussion

The results of this project confirm that Combined Heat and Power (CHP) systems can substantially improve overall energy efficiency compared with separate generation of heat and electricity. When both electricity and useful heat are recovered from the same fuel input, total efficiencies above 70% are achievable, whereas conventional power-only plants often convert only 30–40% of the fuel energy into electricity and reject the remaining heat to the environment. The simple case study of a university building showed that, under reasonable assumptions, a gas-engine-based CHP unit can reduce primary fuel consumption by more than 40% relative to the reference boiler–plus–grid configuration.

The comparison of different prime mover technologies indicates that the optimal CHP solution is strongly site dependent. Reciprocating engines are attractive for small and medium installations due to their high electrical efficiency and moderate cost, while gas and steam turbines are more suitable for larger industrial or district heating applications. Microturbines and fuel cells offer interesting options for distributed generation and low-emission systems but typically require higher capital investment. In all cases, appropriate sizing of the CHP unit to the base thermal load remains a key design guideline.

Beyond conventional cogeneration, the project highlighted the potential benefits of combining CHP with thermal energy storage using Phase Change Materials (PCMs). Integrating a solar flat-plate collector field and a PCM storage tank with a CHP unit could allow part of the thermal demand to be met by renewable energy while the CHP operates closer to its optimum efficiency point. In parallel, the concept of a stand-alone electric PCM storage heater with an external coil offers a practical solution for residential and commercial buildings, enabling load shifting and improved utilisation of cheap or renewable electricity. Overall, CHP and PCM-based storage appear to be complementary technologies that, when combined, can enhance efficiency, flexibility, and sustainability of building energy systems.

9 Conclusion and Recommendations

This course project has presented a structured engineering study of Combined Heat and Power (CHP) systems. Fundamental principles, performance indicators, and typical technologies were reviewed, and a flow diagram of a generic CHP plant was developed to illustrate the main components and energy pathways. A simplified case study of a university building showed that, under reasonable assumptions, a gas-engine-based CHP unit can reduce primary fuel consumption by more than 40% compared with separate generation of heat and electricity.

Beyond conventional cogeneration, the project also explored potential hybrid configurations that combine CHP with solar flat-plate collectors and Phase Change Material (PCM) thermal storage, as well as a stand-alone electric PCM storage heater concept. These ideas build directly on the author's previous senior design project and indicate promising directions for further improving efficiency, flexibility, and the share of renewable energy in building energy systems.

Based on the findings of this work, the following recommendations can be made:

- CHP should be considered as a priority option for facilities with continuous or long-duration heating demands, such as universities, hospitals, hotels, and industrial plants.
- Proper sizing of the CHP unit is critical; designing the system to cover the base thermal load generally leads to high operating hours and good economic performance.
- Future developments should focus on integrating CHP with renewable energy sources and thermal storage—in particular solar thermal collectors and PCM tanks—to further improve sustainability and resilience.
- For academic work, extended modelling and experimental studies of hybrid CHP–solar–PCM systems and electric PCM heaters would provide valuable insights and can serve as a strong foundation for senior design projects or master theses.

Overall, CHP technology offers a practical and mature pathway toward higher energy efficiency and lower emissions. When complemented by solar energy and advanced thermal storage, it has the potential to play an important role in a future low-carbon, distributed energy system.

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